



Original Article

Delayed SINR-Feedback Power Control for High Mobility UAV Communications: A Reliable and Fair OTFS Approach

Submission ID c491a0a6-d55b-40c2-8f27-6ed9ec613d3d

Submission Version Initial Submission

PDF Generation 04 Apr 2026 09:58:58 EST by Atypon ReX

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Additional Information

EDICS

CL1.2 Wireless Communications
CL1.2 Wireless Communications / CL1.2.0 Wireless Techniques and Fading
CL1.2.0 Wireless Techniques and Fading

CL1.2 Wireless Communications / CL1.2.3 Radio Communications
CL1.2.3 Radio Communications

Key Words

Handheld wireless communication
Millimeter wave communication
Millimeter wave radio communication
Mobile communication

Was this manuscript previously rejected?

N/A

Previously rejected in another journal

N/A

Files for peer review

All files submitted by the author for peer review are listed below. Files that could not be converted to PDF are indicated; reviewers are able to access them online.

Name	Type of File	Size	Page
main_paper.pdf	Main Document - PDF	245.5 KB	Page 4

Delayed SINR-Feedback Power Control for High Mobility UAV Communications: A Reliable and Fair OTFS Approach

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Abstract—This letter investigates delayed signal-to-interference-plus-noise ratio (SINR)-feedback-based power control for reliable and fair orthogonal time frequency space (OTFS) unmanned aerial vehicle (UAV) communications under high mobility. Rather than relying on instantaneous channel state information, the proposed scheme updates the transmit power vector using delayed SINR observations, which is well suited to rapidly varying aerial channels. OTFS is shown to provide a significantly smoother effective channel-gain evolution than conventional orthogonal frequency division multiplexing, making delayed feedback more effective for control adaptation. Leveraging this property, a low-complexity controller is developed to enhance weak-user protection and fairness. Numerical results demonstrate that the proposed OTFS-based scheme improves the average minimum SINR and Jain's fairness index as UAV speed increases, while also achieving strong 5th-percentile SINR performance across a range of signal-to-noise ratio conditions. A modest throughput trade-off is observed, but remains acceptable within the targeted reliability- and fairness-oriented operating regime.

Index Terms—UAV communications, SINR-feedback, power control, orthogonal time frequency space (OTFS), reliability, fairness, and high-mobility.

I. INTRODUCTION

Unmanned aerial vehicle (UAV) communications are expected to play an increasingly important role in future wireless networks due to their flexible deployment, strong line-of-sight connectivity, and broad range of applications, including surveillance, emergency response, data collection, and temporary coverage enhancement [1]. However, fast UAV motion induces rapidly time-varying channels, pronounced Doppler effects, and non-negligible feedback delays, making accurate and timely channel state information (CSI) acquisition challenging

in practice [2], [3]. Under such conditions, conventional CSI-dependent adaptation may rely on outdated channel information and thus become unreliable in high-mobility UAV links, particularly when control or resource-allocation loops require frequent channel updates.

Orthogonal time frequency space (OTFS) modulation has emerged as a promising waveform for high-mobility communications, as it operates in the delay-Doppler domain, where the channel representation is more structured and often more stable than in conventional orthogonal frequency division multiplexing (OFDM) systems [4], [5]. This property is particularly advantageous for UAV scenarios, where channel variations are rapid and the usefulness of instantaneous feedback degrades rapidly. Recent OTFS research has established strong foundations in waveform design, signal processing, and transceiver development, while multi-input multi-output (MIMO)-OTFS has further demonstrated its potential in high-Doppler environments [5], [6]. UAV-oriented OTFS studies have also begun to emerge, particularly in mobility-aware link management and highly dynamic aerial networking scenarios [7], [8]. Nevertheless, a clear gap remains in delayed-feedback power control tailored to reliable and fair UAV communications under high mobility, where the focus is on weak-user protection and balanced performance rather than aggressive throughput maximization.

A delayed SINR-feedback approach is attractive in this context because signal-to-interference-plus-noise ratio (SINR) measurements are significantly easier to obtain and feed back than full instantaneous CSI [9]. The main challenge is that delayed feedback rapidly loses relevance as channel dynamics intensify. OTFS is particularly appealing in this regard, as its smoother effective channel-gain evolution allows delayed SINR observations to remain informative for control adaptation compared to OFDM-based systems [4], [7]. Despite this advantage, reliability- and fairness-oriented delayed-feedback power control for OTFS UAV links has received considerably less attention than waveform design, detection, channel estimation, and robust precoding in the existing literature [6], [10].

Motivated by these considerations, this letter is the first to investigate delayed SINR-feedback-based power control for reliable and fair OTFS UAV communications under high mobility. A low-complexity controller is developed to exploit delayed SINR observations without requiring explicit instantaneous CSI. Unlike throughput-driven designs, the proposed approach emphasizes weak-user protection and fairness under fast channel variation and feedback delay. Numerical results demonstrate that, in the considered UAV setting, OTFS yields

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a smoother effective channel-gain evolution than OFDM. Moreover, the proposed controller improves the minimum SINR, enhances Jain's fairness index, and maintains strong tail-SINR performance as mobility increases, with only a modest throughput loss relative to equal power allocation.

II. SYSTEM MODEL

We consider a downlink UAV communication system in which a base station (BS) serves K UAV terminals over an OTFS air interface. For brevity, let $\mathcal{K} = \{1, 2, \dots, K\}$ denote the sets of UAVs. In the OTFS domain, the effective channel gain between the BS and UAV k at time slot t is modeled as [11]

$$g_k(t) = \frac{G_0 |h_k(t)|^2}{(d_k(t))^\nu \psi_k(t)}, \quad (1)$$

where G_0 is the reference channel gain at unit distance, ν is the large-scale path-loss exponent, while $h_k(t)$ and $d_k(t)$ denote the small-scale fading channel coefficient and the propagation distance between the BS and UAV k at time slot t , respectively. We assume that $h_k(t)$ follows Rician fading, reflecting the dominant line-of-sight characteristic commonly observed in UAV air-to-ground links [2], [7]. Moreover, $\psi_k(t)$ captures the residual mobility-induced impairment after OTFS processing of the corresponding channel, and is given by [4]

$$\psi_k(t) = 1 + \kappa_O \left(\frac{f_{D,k}(t)}{f_{D,\text{ref}}} \right)^2. \quad (2)$$

Here, $f_{D,k}(t) \triangleq \frac{v_k(t)}{c} f_c$ denotes the Doppler shift of UAV k at time slot t , where $v_k(t)$ is the UAV velocity, c is the speed of light, and f_c is the carrier frequency. The parameter $f_{D,\text{ref}}$ is a reference Doppler frequency, and κ_O is the OTFS mobility-sensitivity coefficient.

The received signal at UAV k at time slot t is expressed as

$$y_k(t) = \sqrt{\xi_{k,k} P_k(t) g_k(t) x_k(t)} + \sum_{j \in \mathcal{K} \setminus k} \sqrt{\xi_{k,j}(t) P_j(t) g_j(t) x_j(t)} + n_k(t), \quad (3)$$

where $P_k(t)$ denotes the transmit power allocated to UAV k at time slot t , and $x_k(t)$ is a scalar complex baseband transmitted signal with zero mean and unit variance, i.e., $\mathbb{E}\{|x_k(t)|^2\} = 1, \forall k, t$. The noise term $n_k(t) \sim \mathcal{CN}(0, \sigma^2)$ denotes zero-mean Gaussian noise with variance σ^2 , and $\xi_{k,j}(t)$ models the effective interference coupling between UAVs k and j . To satisfy the maximum total power constraint $P_{\max}$ and the minimum per-UAV power constraint $P_{\min}$, the transmit powers $P_k(t), \forall k$, must satisfy

$$\sum_{k \in \mathcal{K}} P_k(t) \leq P_{\max}, \quad P_k(t) \geq P_{\min}, \quad \forall k. \quad (4)$$

Based on (3), the instantaneous received SINR is given by

$$\gamma_k(t) = \frac{\xi_{k,k} P_k(t) g_k(t)}{\sum_{j \in \mathcal{K} \setminus k} \xi_{k,j}(t) P_j(t) g_j(t) + \sigma^2}. \quad (5)$$

Moreover, we adopt a compact geometry- and mobility-aware abstraction in which the coupling depends on the angular separation between UAVs and the average mobility level of the swarm. Let $\theta_k(t)$ denote the angular position of UAV k with respect to the BS, and define the angular separation between UAV k and j as

$$\Delta_{k,j}(t) = \min(|\theta_k(t) - \theta_j(t)|, 2\pi - |\theta_k(t) - \theta_j(t)|). \quad (6)$$

The resulting interference coupling coefficient is modeled as

$$\xi_{k,j}(t) = \begin{cases} 1, & j = k, \\ \xi_0 \left(1 + \rho_v \frac{\bar{v}(t)}{v_{\text{ref}}}\right) \exp\left(-\frac{\Delta_{k,j}(t)}{\phi_0}\right), & j \neq k, \end{cases} \quad (7)$$

where ξ_0 is the baseline interference level, ϕ_0 is the angular decay parameter, ρ_v is the mobility sensitivity factor, $\bar{v}(t) \triangleq \frac{1}{K} \sum_{i=1}^K v_i(t)$ is the average UAV speed, and v_{ref} is a reference speed. Smaller angular separation results in stronger coupling, whereas higher average UAV speed increases the effective interference level under high-mobility conditions. Accordingly, the achievable rate of UAV k at time slot t is given by

$$R_k(t) = \log_2(1 + \gamma_k(t)). \quad (8)$$

To evaluate how evenly the achieved rates are distributed across UAVs, fairness is quantified using Jain's fairness index [12]

$$J(t) = \frac{\left(\sum_{j \in \mathcal{K}} R_k(t)\right)^2}{K \sum_{j \in \mathcal{K}} R_k^2(t)}. \quad (9)$$

A value of $J(t)$ closer to one indicates a more balanced rate distribution among the UAVs.

Instead of assuming full and instantaneous CSI at the BS, we consider a delayed SINR-feedback model. Specifically, the BS receives from UAV k a delayed SINR observation

$$\gamma_k^{\text{fb}}(t) = \gamma_k(t - d), \quad (10)$$

where $d \geq 0$ denotes the feedback delay measured in time slots. Due to rapid channel variations in UAV links, feedback delay generally reduces the relevance of conventional instantaneous link indicators. However, when OTFS induces a smoother effective channel evolution, delayed SINR observations can remain more informative for control adaptation [4], [7]. The objective is to design a low-complexity delayed-feedback power controller that improves the minimum SINR and fairness under high mobility, while satisfying the constraints in (4). Direct optimization is challenging because the UAVs are interference-coupled and the BS observes only delayed SINR feedback. Therefore, we adopt a feedback-control perspective and design a mapping $\mathcal{F}(\cdot)$ from delayed SINR observations to the next-slot transmit powers.

The proposed design targets reliable and fair UAV communications under delayed SINR feedback while maintaining acceptable throughput. A utility function that captures this tradeoff is defined as

$$U(t) = \omega_1 \gamma_{\min}(t) + \omega_2 J(t) + (1 - \omega_1 - \omega_2) \sum_{k \in \mathcal{K}} R_k(t), \quad (11)$$

where $\gamma_{\min}(t) \triangleq \min_{k \in \mathcal{K}} \gamma_k(t)$, $J(t)$ is Jain's fairness index, and $\omega_1, \omega_2 \in [0, 1]$, satisfying $\omega_1 + \omega_2 \leq 1$. The sum-rate term reflects overall spectral-efficiency, while $\gamma_{\min}(t)$ explicitly protects the weakest UAV, whose performance may otherwise be masked by the aggregate rate. The fairness term $J(t)$ further discourages highly uneven rate allocation across UAVs. Larger values of ω_1 and ω_2 therefore place greater emphasis on weak-user protection and fairness, respectively.

Directly maximizing (11) is challenging due to interference coupling, delayed SINR observations, and the dependence of

feedback usefulness on channel dynamics under high mobility. Therefore, this work adopts a feedback-control perspective, and the power-update problem is formulated as the design of a low-complexity mapping

$$\mathbf{P}(t+1) = \mathcal{F}(\mathbf{P}(t), \gamma^{\text{fb}}(t)), \quad (12)$$

where $\mathbf{P}(t) \triangleq [P_1(t), P_2(t), \dots, P_K(t)]^T$ and $\gamma^{\text{fb}}(t) \triangleq [\gamma_1^{\text{fb}}(t), \gamma_2^{\text{fb}}(t), \dots, \gamma_K^{\text{fb}}(t)]^T$ represent the transmit power vector and the delayed SINR feedback vector, respectively. The mapping $\mathcal{F}(\cdot)$ is designed to exploit delayed SINR observations via prediction, smoothing, reliability-oriented UAV scoring, and projected multiplicative power updates, while satisfying the power constraints in (4).

III. PROPOSED DELAYED SINR-FEEDBACK POWER CONTROL

Motivated by the smoother effective gain evolution offered by OTFS in high-mobility channels [4], [13], the proposed delayed SINR-feedback controller aims to improve reliability and fairness rather than purely maximizing instantaneous throughput. Instead of explicitly solving (11) at each time slot, the BS updates the power vector using a low-complexity prediction-correction rule driven solely by delayed SINR observations, without requiring explicit instantaneous CSI.

To partially compensate for the feedback delay, the BS forms a predicted SINR as

$$\hat{\gamma}_k(t) = \left[\gamma_k^{\text{fb}}(t) + \frac{\lambda}{\bar{d}} (\gamma_k^{\text{fb}}(t) - \gamma_k^{\text{fb}}(t-1)) \right]_{\gamma_{\text{lb}}}^{\gamma_{\text{ub}}}, \quad (13)$$

where $\lambda \geq 0$ is the delay-compensation factor, $\bar{d} = \max(d, 1)$ is introduced to avoid division by zero in the no-delay case, and $[\cdot]_{\gamma_{\text{lb}}}^{\gamma_{\text{ub}}}$ denotes clipping onto a prescribed SINR interval to prevent unrealistically small or large predicted values.

To suppress short-term fluctuations, the controller computes a smoothed SINR estimate

$$\bar{\gamma}_k(t) = \alpha \bar{\gamma}_k(t-1) + (1-\alpha) \hat{\gamma}_k(t), \quad (14)$$

where $\alpha \in [0, 1]$ is the smoothing coefficient. Larger α yields a more conservative control response, whereas smaller α makes the controller more responsive to recent feedback. To protect vulnerable users, the controller measures the SINR shortfall relative to the target reliability threshold γ_{th} as

$$s_k(t) = \max(\gamma_{\text{th}} - \bar{\gamma}_k(t), 0). \quad (15)$$

A positive $s_k(t)$ indicates that UAV k is below the desired SINR level and requires additional protection. When at least one UAV experiences a nonzero shortfall, the control priority is determined by the normalized shortfall

$$q_k(t) = \frac{s_k(t)}{\sum_{i \in \mathcal{K}} s_i(t)}, \text{ if } \sum_{i \in \mathcal{K}} s_i(t) > 0. \quad (16)$$

In contrast, when all UAVs satisfy the target threshold, the controller switches to a mild gain-aware balancing mode to avoid unnecessary power concentration. Let $g_k^{\text{fb}}(t)$ denote a delayed link-quality proxy available at the BS. The corresponding normalized score is then given by

$$q_k(t) = \frac{g_k^{\text{fb}}(t)}{\sum_{i \in \mathcal{K}} g_i^{\text{fb}}(t)}, \text{ if } \sum_{i \in \mathcal{K}} s_i(t) = 0. \quad (17)$$

To ensure that the subsequent power update redistributes power across UAVs, rather than merely scaling all entries in the same direction, the controller uses a centered score

$$\tilde{q}_k(t) = q_k(t) - \frac{1}{K} \sum_{i \in \mathcal{K}} q_i(t). \quad (18)$$

The power update is performed in the log-power domain as

$$u_k(t) = \mu_t \tilde{q}_k(t) + \beta (\tilde{q}_k(t) - \tilde{q}_k(t-1)), \quad (19)$$

where μ_t is the effective step size and $\beta \geq 0$ is a momentum coefficient. To improve robustness under larger delays, the step size is adapted as

$$\mu_t = \frac{\mu}{1 + \eta d}, \quad (20)$$

where $\mu > 0$ is the nominal step size and $\eta \geq 0$ controls the delay sensitivity. To avoid excessive slot-to-slot power variations, $u_k(t)$ is clipped as

$$u_k(t) \leftarrow \min(\max(u_k(t), -\Delta_{\text{max}}), \Delta_{\text{max}}), \quad (21)$$

where $\Delta_{\text{max}} > 0$ denotes the maximum log-power increment. This clipping limits instantaneous aggressiveness and helps maintain stable power adaptation under delayed feedback.

The tentative power allocation is then computed as

$$\tilde{P}_k(t+1) = P_k(t) \exp(u_k(t)). \quad (22)$$

Since (22) may violate the power constraints in (4), the vector $\tilde{\mathbf{P}}(t+1) \triangleq [\tilde{P}_1(t), \tilde{P}_2(t), \dots, \tilde{P}_K(t)]^T$ is projected onto the feasible set

$$\mathcal{P} = \left\{ \mathbf{P} \in \mathbb{R}_+^K : \sum_{k \in \mathcal{K}} P_k \leq P_{\text{max}}, P_k \geq P_{\text{min}}, \forall k \right\}, \quad (23)$$

where $\mathbb{R}_+^K$ denotes the K -dimensional nonnegative orthant. Accordingly, the final update is given by

$$\mathbf{P}(t+1) = \Pi_{\mathcal{P}}(\tilde{\mathbf{P}}(t+1)), \quad (24)$$

where $\Pi_{\mathcal{P}}(\cdot)$ denotes Euclidean projection onto $\mathcal{P}$. Compared with slot-wise CSI-based optimization, the proposed method requires only delayed SINR feedback, a lightweight prediction step, and a simple projection. Its low computational complexity and robustness to fast channel variation make it well suited for highly mobile UAV links. The overall algorithm of the proposed delayed SINR-feedback power control is summarized in Algorithm 1.

The proposed controller has low per-slot complexity. SINR prediction, smoothing, shortfall evaluation, score construction, and log-domain power updates each require $\mathcal{O}(K)$ floating point operations (FLOPs), while the dominant projection step requires $\mathcal{O}(K \log K)$ FLOPs under a standard bounded-simplex implementation. The overall per-slot complexity is therefore $\mathcal{O}(K \log K)$ FLOPs, with memory requirement $\mathcal{O}(K)$.

IV. SIMULATION RESULTS

In this section, we perform system-level simulations to evaluate the performance of the proposed method. The BS is located at the center of the network, and K UAVs are uniformly initialized at random such that the BS-UAV distance lies within 120–300 m. Unless otherwise specified, the system parameters are set to $K = 6$, $P_{\text{max}} = 1$, $P_{\text{min}} = 0.02$, and $f_c = 5.8$ GHz. The power quantities are normalized with respect to the noise power, which is set to $\sigma^2 = 1$ in the simulations. Accordingly, the average operating signal-to-noise

Algorithm 1: Proposed Delayed SINR-Feedback Power Control

Input: $P_{\max}, P_{\min}, \gamma_{\text{th}}, \bar{d}, \Delta_{\max}, \mu, \beta, \eta, \lambda, \alpha, \{\gamma_k^{\text{fb}}(0)\}_{\forall k \in \mathcal{K}}$.

- 1 Initialize: $P_k(1) = P_{\max}/K, \bar{\gamma}_k(0) = \gamma_k^{\text{fb}}(0), \tilde{q}_k(0) = 0, \forall k$;
- 2 **for** $t = 1, 2, \dots, T - 1$ **do**
- 3 Receive delayed SINR feedbacks $\gamma_k^{\text{fb}}(t), \forall k$;
- 4 Compute predicted SINRs $\hat{\gamma}_k(t)$ by (13), $\forall k$;
- 5 Update smoothed SINRs $\bar{\gamma}_k(t)$ by (14), $\forall k$;
- 6 Compute shortfalls $s_k(t)$ using (15), $\forall k$;
- 7 **if** $\sum_{k \in \mathcal{K}} s_k(t) > 0$ **then**
- 8 Set $q_k(t)$ according to (16), $\forall k$;
- 9 **end**
- 10 **else**
- 11 Obtain delayed gain proxy $g_k^{\text{fb}}(t)$ and set $q_k(t)$ according to (17), $\forall k$;
- 12 **end**
- 13 Center the scores $\tilde{q}_k(t)$ using (18), $\forall k$;
- 14 Compute delay-aware step size μ_t using (20);
- 15 Update signals $u_k(t)$ according to (19) and clip it using (21), $\forall k$;
- 16 Update tentative powers $\tilde{P}_k(t+1)$ using (22), $\forall k$;
- 17 Project $\tilde{P}_k(t+1)$ onto the feasible set using (23), $\forall k$;
- 18 **end**

ratio (SNR) of 10 dB is defined relative to this normalized noise level. Instead of assuming instantaneous CSI at the BS, the controller operates only on delayed SINR feedback, where the SINR measured at the UAV is fed back to the BS after $d = 2$ slots and used for power adaptation in subsequent transmissions. The coefficients ω_1 and ω_2 are tuned offline over the feasible set $\{\omega_1, \omega_2 \in [0, 1] : \omega_1 + \omega_2 \leq 1\}$, and the best pair used in the simulations is $(\omega_1, \omega_2) = (0.6, 0.4)$. The mobility-dependent channel follows a Rician factor of 7 dB, path-loss exponent $\nu = 2.2$, OTFS mobility-sensitivity coefficient $\kappa_0 = 0.35$, reference Doppler frequency $f_{D,\text{ref}} = 200$ Hz, mobility sensitivity factor $\rho_v = 0.80$, angular decay parameter $\phi_0 = 25^\circ$, reference speed $v_{\text{ref}} = 100$ m/s, and reference channel gain $G_0 = 1$ [11]. The baseline interference level is set to $\xi_0 = 0.05$, which corresponds to a weak normalized residual coupling regime. This choice aligns with geometry-dependent UAV interference models, where the interference level is strongly influenced by directional separation and propagation conditions [11]. The default UAV speed is $v = 40$ m/s. The proposed delayed-feedback controller uses the parameter set $\lambda = 0.60, \alpha = 0.60, \mu = 0.14, \beta = 0.10$, and $\eta = 0.25$. The reliability threshold is set to $\gamma_{\text{th}} = -3$ dB, and the update signal is clipped with a maximum log-power increment of $\Delta_{\max} = 0.18$. These values are selected empirically from a stable operating region to mitigate oscillatory projected updates while preserving improvements in minimum-SINR and fairness under delayed feedback.

The proposed OTFS-based controller is compared with several benchmark schemes, including equal power allocation,

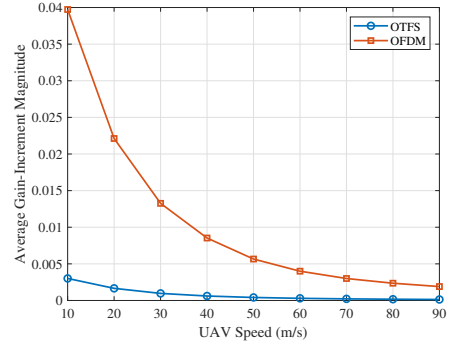


Fig. 1. Average gain-increment magnitude versus UAV speed for OTFS and OFDM.

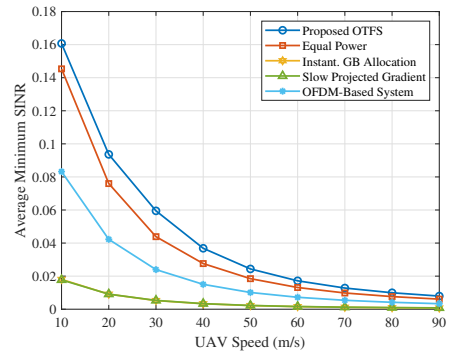


Fig. 2. Average minimum SINR versus UAV speed.

instantaneous gain-based allocation, a slow projected-gradient method, and an OFDM-based delayed-feedback controller using the same update structure. The goal of this comparison is not to claim universal superiority across all metrics, but rather to assess whether OTFS-enabled delayed-feedback control improves low-end SINR performance, fairness, and robustness under high mobility.

Fig. 1 compares the average gain-increment magnitude of OTFS and OFDM as the UAV speed increases. As observed, OTFS consistently exhibits smaller gain-increment magnitudes than OFDM across the entire speed range, indicating that the effective OTFS link evolves more smoothly under high mobility. The performance gap remains pronounced even in the high-speed regime, where Doppler effects are stronger and conventional feedback becomes less reliable. This smoother gain evolution makes delayed SINR feedback more informative and better suited for control adaptation, as past observations remain more relevant to current channel conditions. This behavior provides direct physical justification for using OTFS in delayed-feedback power control for fast-varying UAV links.

Fig. 2 shows the average minimum SINR as a function of UAV speed. As mobility increases, all schemes experience performance degradation due to faster channel variation and reduced feedback reliability. Nevertheless, the proposed OTFS-based controller remains highly competitive in the high-speed regime and provides stronger protection for weak UAVs than the benchmark schemes. While equal power allocation can perform reasonably well at low speeds, the proposed method becomes increasingly advantageous as mobility increases and

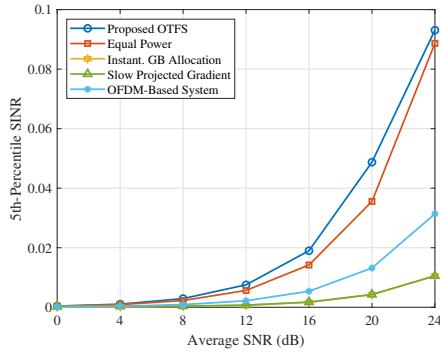


Fig. 3. 5th-percentile SINR versus average SNR.

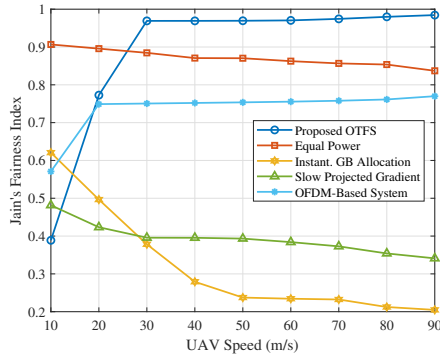


Fig. 4. Jain's fairness index versus UAV speed.

delayed feedback becomes more challenging to exploit.

Fig. 3 presents the 5th-percentile SINR versus the average SNR. The proposed OTFS-based scheme achieves strong tail-SINR performance, particularly in the moderate-to-high SNR regime where weak-UAV reliability becomes a key differentiator. The improvement in the lower tail indicates that the delayed-feedback controller effectively protects vulnerable users, rather than merely improving average link quality. This is merely relevant in UAV systems with rapid channel variation, where outdated feedback can significantly degrade low-end performance. A modest throughput loss relative to more aggressive baselines is acceptable under the targeted reliability- and fairness-oriented design.

Fig. 4 reports Jain's fairness index versus UAV speed. The proposed OTFS controller achieves the highest or near-highest fairness across the entire speed range, with more pronounced gains in the medium-to-high mobility regime where channel variation is faster and delayed feedback is less reliable. This trend indicates that the smoother effective gain evolution under OTFS allows the delayed SINR-feedback loop to remain sufficiently informative for balanced power adaptation, even at high speeds. Consequently, the proposed controller avoids excessive power concentration on a small subset of favorable UAVs and instead maintains a more uniform rate distribution across. While equal power allocation can achieve relatively high fairness in some regimes, the proposed method attains comparable or better balance while being explicitly incorporating delayed SINR information and weak-user protection. The observed fairness improvement is consistent with the reliability-oriented design objective, where UAVs with larger

delayed-SINR shortfalls are prioritized during power updates.

V. CONCLUSION

This letter proposes a delayed SINR-feedback power control framework for reliable and fair OTFS UAV communications under high mobility. The proposed design relies solely on delayed SINR observations and exploits the smoother effective gain evolution of OTFS to enable more effective feedback-driven adaptation in fast-varying channels. Numerical results show that exhibits significantly smaller gain variations than OFDM, while the proposed controller improves the average minimum SINR, maintains a high Jain's fairness index across the UAV-speed range, and achieves strong 5th-percentile SINR performance over SNR. A modest loss in average throughput relative to more throughput-oriented baselines is consistent with the design objective of protecting weak UAVs and balancing performance under delayed feedback. Future work may extend the proposed framework to MIMO-OTFS UAV systems and investigate joint optimization of waveform design, beamforming, and feedback-control mechanisms.

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